

Assessment Brief Template

Academic Year	2024-25
Semester	Y1 (Semester 01)
Module Number	CM1601
Module Title	Programming Fundamentals
Assessment Method	Course Work
Deadline (time and date)	17th April 2025
Submission	Assessment Dropbox in the Module Study Area in Campus Moodle.
Word Limit (See Assessment Word Limit Statement)	
Module Co-ordinator	Dinusha Ruwan Kumara

What knowledge and/or skills will I develop by undertaking the assessment?

1. Demonstrate competence in the algorithmic approach to problem solving.
2. Design, code, compile, test and execute fundamental programming concepts using a high-level programming language.
3. Construct robust, maintainable programs that use object-orientated analysis and design principles.

On successful completion of the assessment students will be able to achieve the following Learning Outcomes:

1. Design, code, compile, test and execute fundamental programming concepts using a high-level programming language.
2. Construct robust, maintainable programs that use object-orientated analysis and design principles.

Please also refer to the Module Descriptor, available from the module Moodle study area.

Task(s) - content

Make explicit reference to the relevance of the material content each week to the learning outcomes and the assessment task(s).

Objective

You are instructed to create a command line application for the following scenario using Python.

Internet Cafes have become a popular business due to rapid technological changes and higher demand. Many people, especially students who do not have access to a personal computer, visit internet cafes to complete their projects and assignments, as well as for entertainment.

David is planning to start an Internet cafe named 'One Net Cafe' in his hometown so that people without personal computers can use the facilities. He has bought some of the items needed for the cafe and is looking for suppliers to purchase the rest. He has already received details of six wholesale suppliers and each one sells good quality items at low prices. Upon researching, the quotations were found to be almost the same. Due to the similarities among the suppliers, David has decided to select four dealers randomly to purchase the remaining products. As a high number of equipment items are required for the café, David needs a system to manage inventory.

The system should display the following console menu when launched.

- Type **AID** for adding item details.
- Type **DID** for deleting item details.
- Type **UID** for updating item details.
- Type **VID** for viewing the items table. (Sort according to the items category) and print the current total.
- Type **SID** for saving the item details to the text file at any time.
- Type **SDD** for selecting four dealers randomly from a file.
- Type **VRL** for displaying all the details of the randomly selected dealers. (Sorted according to the location.)
- Type **LDI** for displaying the items of the given dealer.
- Type **ESC** to exit the program.

Coursework Sections	Detailed content	Learning Outcome	Topic/ Week /Module Activity
AID- Adding item details	The system should allow the owner to add the details of items to his system by prompting the required information. Item Code, Item Name, Item Brand, Price, Quantity, Category, Purchased Date. example: 001, Speaker, Samsung, Rs 21966.00, 10, Speakers, 2023/02/07	1	Lecture 2 – Basic Built in Functions Lecture 7 - Functions

What is expected of me in this assessment?

DID -Deleting item details	The system should allow the owner to delete details of an item, searching by item code.	1	Lecture 3 - Python Decision Making
UID -Updating item details	System should allow the owner to update details of an item, searching by item code.	1	Lecture 3 - Python Decision Making
VID-Viewing the items table and current total	The system should display the item details in descending order with considering item id and print the current total of the items that are already purchased. This table should be formatted neatly with all the details.	1	Lecture 7: Python String
SID- Saving the item details to the text file	The system should allow the owner to save the items' details to the same text file at any point. (When an item added, deleted, and updated.)	1,2	Lecture 10 – File handling
SDD - Selecting four dealers randomly from a file	The system should simulate a random draw and select four dealers accordingly from a text file. Once selection is done display the message '4 Dealers are Selected Randomly'. A dealer would have the following details. Dealers' Name, Contact No, Location, items (3 items for each dealer). Item details should include Name, Brand, Price, Quantity. (Text file(s) should be created by yourself accordingly)	1,2	Lecture 10 – File handling Lecture 5 – Python Loops II Lecture 6- Python Exception Handling
VRL - Displaying all the details of the randomly selected dealers	The system should display all the details of the randomly selected dealers and be sorted according to the location. Use your own algorithm for sorting.	1	Lecture 3 - Python Decision Making Lecture 4 – Python Loops I
LDI - Display the items of the given dealer	The system should be able to view the items when the name of one of the randomly selected dealers is given.	1	Lecture 8- Python Collections

What is expected of me in this assessment?

Task(s) - format

1. All the functionalities mentioned above need to be fully implemented. Code should be submitted as a single zip file to the link in assessment section. Check the following key points before you start the implementation.
 - a. A console menu system to enable the user input with validations should be implemented.
 - b. Information entered should be saved in text file(s).
 - c. You are not allowed to use databases for this coursework.
 - d. When sorting data, you are not allowed to use inbuilt python libraries and marks will be given only for your own algorithms.
2. You should submit a report containing your python code for each function with a brief description and a screenshot of the output.

Important: If you are further developing the application or doing any modifications due to a justifiable fact, please mention that in your report.

The approaches taken, assumptions made, will be checked thoroughly during the viva. Full marks will be awarded for each criterion only if you implement successfully and defend well during the viva.

How will I be graded?

A grade will be provided for each criterion on the feedback grid which is specific to the assessment. The overall grade for the assessment will be calculated using the algorithm below.

A	At least 50% of the feedback grid to be at Grade A, at least 75% of the feedback grid to be at Grade B or better, and normally 100% of the feedback grid to be at Grade C or better.
B	At least 50% of the feedback grid to be at Grade B or better, at least 75% of the feedback grid to be at Grade C or better, and normally 100% of the feedback grid to be at Grade D or better.
C	At least 50% of the feedback grid to be at Grade C or better, and at least 75% of the feedback grid to be at Grade D or better.
D	At least 50% of the feedback grid to be at Grade D or better, and at least 75% of the feedback grid to be at Grade E or better.
E	At least 50% of the feedback grid to be at Grade E or better.
F	Failing to achieve at least 50% of the feedback grid to be at Grade E or better.
NS	Non-submission.

Feedback grid

GRADE	A	B	C	D	E	F
DEFINITION / CRITERIA (WEIGHTING)	EXCELLENT Outstanding Performance	COMMENDABLE/VERY GOOD Meritorious Performance	GOOD Highly Competent Performance	SATISFACTORY Competent Performance	BORDERLINE FAIL	UNSATISFACTORY Fail
A console menu system with validations (10x %) Grade:	All menu Options are displayed in a console correctly. Users can select the option by providing an input. Inputs are validated. Students demonstrate excellent knowledge on the task through viva.	More than 7 menu Options are displayed in a console correctly. Users can select the option by providing an input. basic input validation can be seen. Students demonstrate very good knowledge on the task through viva.	More than 6 menu Options are displayed in a console correctly. Users can select many options by providing an input. basic input validation. Students demonstrate good knowledge on the task through viva.	More than 5 menu Options are displayed in a console correctly. Users can select many options by providing an input. minor input validation. Students demonstrate adequate knowledge on the task through viva.	Less than 5 menu options are displayed in the console. Users can only select a few options via input. no input validations can be seen. Student does not demonstrate adequate knowledge on the task through viva.	No console menu to be seen. User inputs are not considered. Demonstrates no knowledge on the task. Program does not run.
Add, Delete & Update item details (10x %) Grade:	Adding, deleting, and updating items details can be done by giving the item code. Duplicate item records are handled, and all the inputs are correctly validated. Sub-menu for each function works flawlessly. Students demonstrate excellent knowledge on the task through viva.	Adding, deleting, and updating items details can be done by a single attribute. duplicate item records are handled, and all the inputs are correctly validated. sub-menu for each function works. Students demonstrate very good knowledge on the task through viva.	Adding, deleting, and updating items details can be done. No duplicate item records are handled. Basic inputs are validated. Sub-menu for each function works. Students demonstrate good knowledge on the task through viva.	Adding, deleting, and updating of items details can be done. No duplicate item records are handled. No inputs are validated. Sub-menu for each function works. Students demonstrate adequate knowledge on the task through viva.	Only two of the functions out of three can be done. No duplicate item records are handled. Basic inputs are not validated. Sub-menu for each function works with errors. Students demonstrate low knowledge on the task through viva.	Functions do not work. Demonstrates no knowledge on the task. Program does not run.

GRADE	A	B	C	D	E	F
DEFINITION / CRITERIA (WEIGHTING)	EXCELLENT Outstanding Performance	COMMENDABLE/VERY GOOD Meritorious Performance	GOOD Highly Competent Performance	SATISFACTORY Competent Performance	BORDERLINE FAIL	UNSATISFACTORY Fail
View Item details and view randomly selected dealer's details (20x %) Grade:	Item details and randomly selected dealers' details are neatly formatted. Item details are sorted according to the item code and dealers' details are sorted according to the location. Student has implemented two sorting algorithms. Total of the current items displayed. All the required columns can be viewed. Demonstrates excellent knowledge on the task through viva.	Item details and randomly selected dealers' details are neatly formatted. Item details are sorted according to the item code and dealers' details are sorted according to the location. Student has implemented two sorting algorithms. All the required columns can be viewed. Total of the current items displayed. Demonstrates very good knowledge on the task through viva.	Item details and randomly selected dealers' details are printed on the console. Item details are sorted according to the item code and dealers' details are sorted according to the location. Student has implemented at least one sorting algorithm. Most of the required columns can be viewed. Total of the current items displayed. Demonstrates good knowledge on the task through viva.	Item details and randomly selected dealers' details are printed on the console. Item details are sorted according to the item code and dealers' details are sorted according to the location. Student has used default sort function. Most of the required columns can be viewed. Total of the current items not calculated and displayed. Demonstrates satisfactory knowledge on the task through viva.	Item details and randomly selected dealers' details are printed on the console with no formatting. No sorting can be seen, or sorting is incorrect. Most of the required columns are missing in the output. Total of the current items not calculated and displayed. Demonstrates low knowledge on the task through viva.	Item details and randomly selected dealers' details are not printed on the console. Total of the current items not calculated and displayed. Demonstrates no knowledge on the task through viva.
File handling (15x %) Grade:	Details of the items are saved to a text file successfully and details of the dealers are taken from a text file(s). Necessary validations are being implemented. Duplicate errors were addressed. Exceptions were handled in the code. Resume capability of the system have been implemented. The student demonstrates excellent knowledge on the task through viva.	Details of the items are saved to a text file successfully and details of the dealers are taken from a text file(s). Necessary validations are being implemented. Duplicate errors were addressed. Exceptions were handled in the code. Resume capability of the system have been implemented. The student demonstrates very good knowledge on the task through viva.	Details of the items are saved to a text file successfully or details of the dealers are taken from a text file(s). Necessary validations are not being implemented. Duplicate errors were addressed. Exceptions were handled in the code. Resume capability of the system have been implemented. The student demonstrates good knowledge on the task through viva.	Details of the items are saved to a text file successfully or details of the dealers are taken from a text file(s). Exceptions were not handled in the code. The student demonstrates satisfactory knowledge on the task through viva.	Attempts were made to code the file handling but not working. File has been created without data. The student demonstrates unsatisfactory knowledge on the task through viva.	File handling is not implemented. student Demonstrates no knowledge on the task through viva.

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Random dealers (15x %) Grade: ☐	Random dealers are selected. Duplicates are being validated. The student demonstrates excellent knowledge on the task through viva.	Random dealers are selected. Duplicates are being validated. The student demonstrates very good knowledge on the task through viva.	Random dealers are selected. Duplicates are being validated. The student demonstrates good knowledge on the task through viva.	Random dealers are selected. No validations can be seen. The student demonstrates satisfactory knowledge on the task through viva.	Random dealers are generated with no relevance to the existing data. No validations can be seen. The student demonstrates unsatisfactory knowledge on the task through viva.	Random dealers' selection is not implemented. Student demonstrates no knowledge on the task through viva.
Display the items of the given dealer (10x %) Grade: ☐	Item details of the given dealer is displayed in the console. The student demonstrates excellent knowledge on the task through viva.	Item details of the given dealer is displayed in the console. The student demonstrates very good knowledge on the task through viva.	Item details of the given dealer is displayed in the console. The student demonstrates very good knowledge on the task through viva.	Item details of the given dealer is displayed in the console. The student demonstrates satisfactory knowledge on the task through viva.	Item details of a dealer is displayed in the console. The student demonstrates unsatisfactory knowledge on the task through viva.	Item details of the given dealer is not displayed in the console. The student demonstrates no knowledge on the task through viva.
Report & Test Plan (10x %) Grade: ☐	A Detailed report can be seen. Report has all the required details. Code is added to the report and clear descriptions are given. All the screenshots of outputs are clearly attached. Excellent formatting can be seen throughout the report.	Report has some required details. Code is added to the report and clear descriptions are given. All the screenshots of outputs are clearly attached. Average formatting can be seen throughout the report.	Report has some required details. Some of the code is added to the report and descriptions are given. Few screenshots are attached. Average formatting can be seen throughout the report.	Report has some required details. Some of the code is added to the report and descriptions are given. No screenshots are attached. Formatting cannot be seen throughout the report.	Incomplete report. Some of the code is added to the report and descriptions are not given clearly. No screenshots are attached.	No report or Incomplete report can be found. No description given. No codes and screenshots attached.

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Code Quality and improvements (10x %) Grade:	Code comments, in code references, Proper naming conventions, Indentation and use of proper coding structures can be seen. Smart Improvements on the given specification is evident.	Code comments, Proper naming conventions, Indentation can be seen. Many Improvements on the given specification is evident.	Code comments, Proper naming conventions, Indentation can be seen. At least one Improvements on the given specification is evident.	Code comments, Proper naming conventions, Indentation can be seen.	No comments to be seen. indentation is not consistent. Major naming errors are visible.	No comments to be seen. No indentation of the code. Has no idea on naming conventions.

Coursework received late, without valid reason, will be regarded as a non-submission (NS) and one of your assessment opportunities will be lost.

What else is important to my assessment?

What is plagiarism?

"Plagiarism is the practice of presenting the thoughts, writings or other output of another or others as original, without acknowledgement of their source(s) at the point of their use in the student's work. All materials including text, data, diagrams, or other illustrations used to support a piece of work, whether from a printed publication or from electronic media, should be appropriately identified and referenced and should not normally be copied directly unless as an acknowledged quotation. Text, opinions or ideas translated into the words of the individual student should in all cases acknowledge the original source" ([RGU 2022](#)).

What is collusion?

"Collusion is defined as two or more people working together with the intention of deceiving another. Within the academic environment this can occur when students work with others on an assignment, or part of an assignment, that is intended to be completed separately" ([RGU 2022](#)).

For further information please see [Academic Integrity](#).

What is the Assessment Word Limit Statement?

It is important that you adhere to the Word Limit specified above. The Assessment Word Limit Statement lists what is included and excluded from the word count, along with the penalty for exceeding the upper limit.

What if I'm unable to submit?

- The University operates a [Fit to Sit Policy](#) which means that if you undertake an assessment then you are declaring yourself well enough to do so.
- If you require an extension, you should complete and submit a [Coursework Extension Form](#). This form is available on the [RGU Student and Applicant Forms](#) page.
- Further support is available from your Course Leader.

What additional support is available?

- [RGU Study Skills](#) provide advice and guidance on academic writing, study skills, maths and statistics and basic IT.
- [RGU Library guidance on referencing and citing](#).
- [The Inclusion Centre: Disability & Dyslexia](#).
- Your Module Coordinator, Course Leader and designated Personal Tutor can also provide support.

What are the University rules on assessment?

The University Regulation '[A4: Assessment and Recommendations of Assessment Boards](#)' sets out important information about assessment and how it is conducted across the University.